# UML Class Diagram for Robotic Path P

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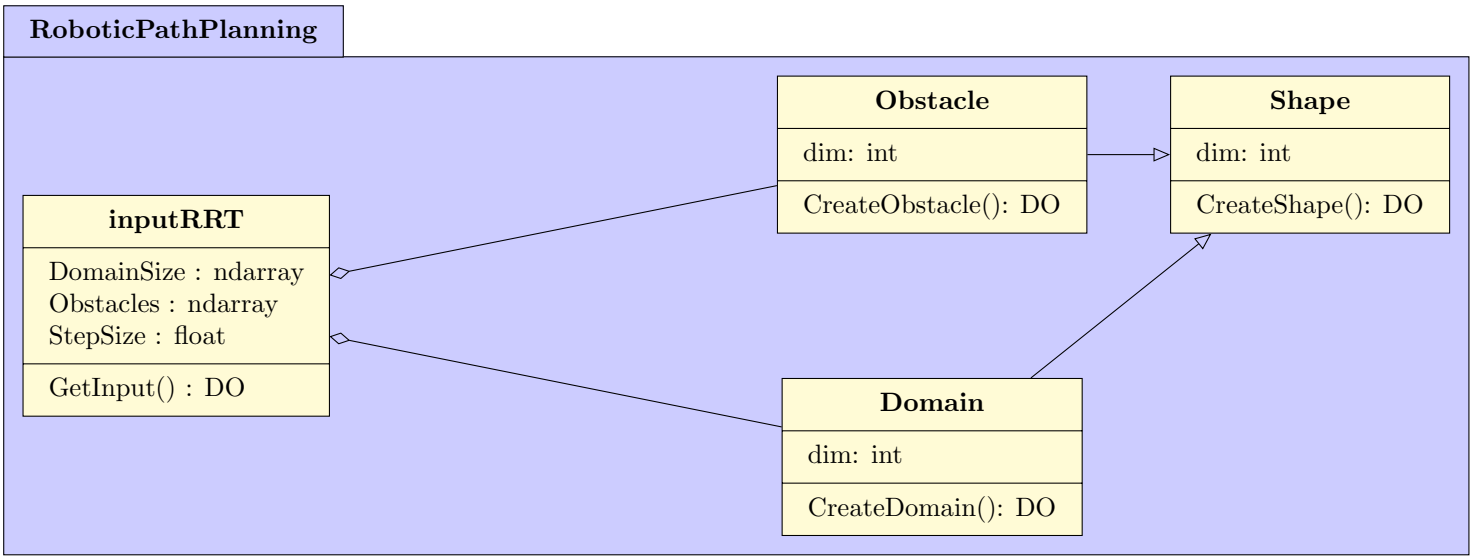

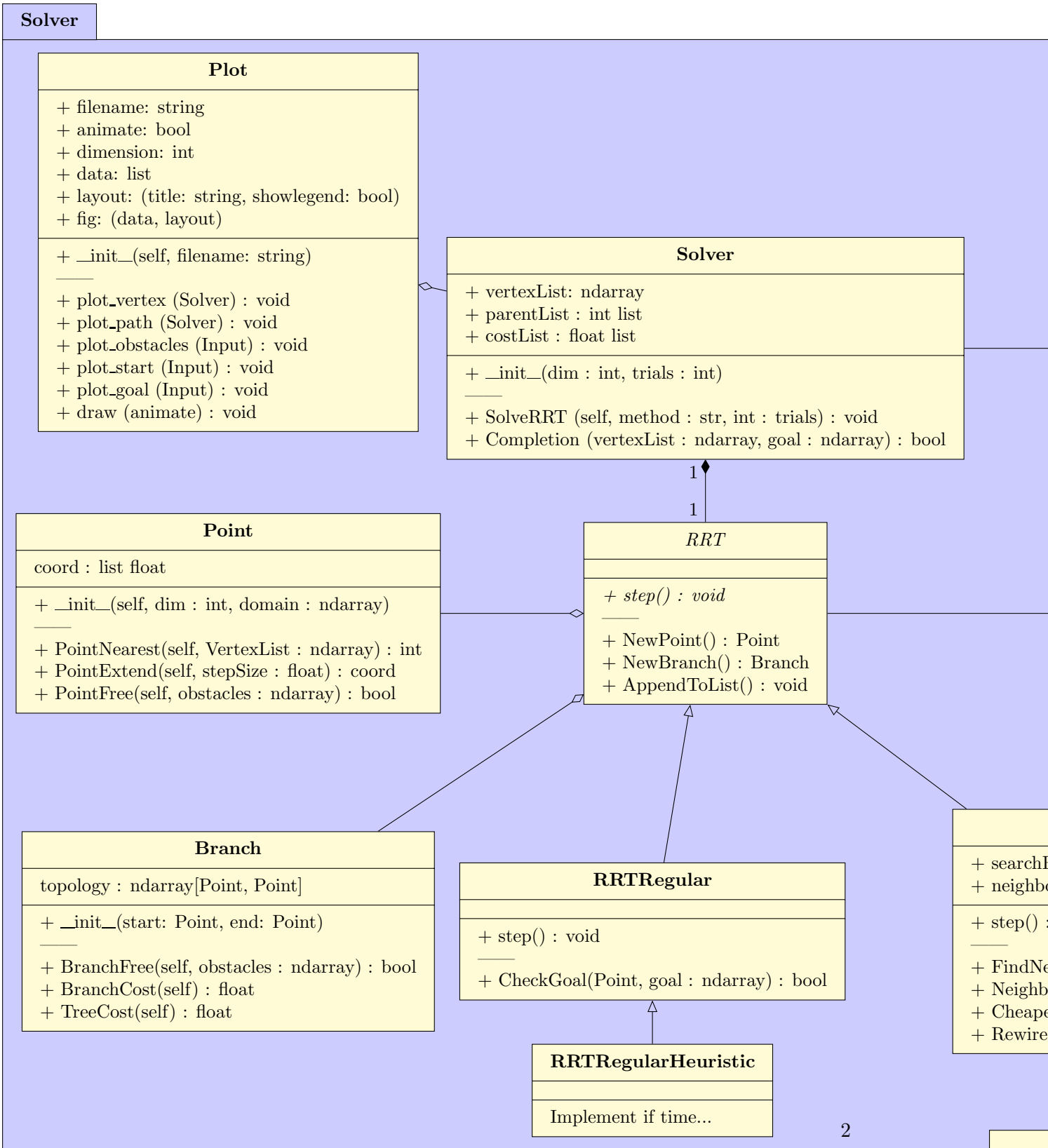
Solver
Plot
+ filename: string
+ animate: bool
+ dimension: int
+ data: list
+ layout: (title: string, showlegend: bool)
+ fig: (data, layout)
+ __init__(self, filename: string)
+ plot_vertex (Solver) : void
+ plot_path (Solver) : void
+ plot_obstacles (Input) : void
+ plot_start (Input) : void
+ plot_goal (Input) : void
+ draw (animate) : void
Solver
+ vertexList: ndarray
+ parentList : int list
+ costList : float list
+ __init__(dim : int, trials : int)
+ SolveRRT (self, method : str, int : trials) : void
+ Completion (vertexList : ndarray, goal : ndarray) : bool
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1
Point
coord : list float
+ __init__(self, dim : int, domain : ndarray)
+ PointNearest(self, VertexList : ndarray) : int
+ PointExtend(self, stepSize : float) : coord
+ PointFree(self, obstacles : ndarray) : bool
RRT
+ step() : void
+ NewPoint() : Point
+ NewBranch() : Branch
+ AppendToList() : void
Branch
topology : ndarray[Point, Point]
+ __init__(start: Point, end: Point)
+ BranchFree(self, obstacles : ndarray) : bool
+ BranchCost(self) : float
+ TreeCost(self) : float
RRTRegular
+ step() : void
+ CheckGoal(Point, goal : ndarray) : bool
RRTRegularHeuristic
Implement if time...